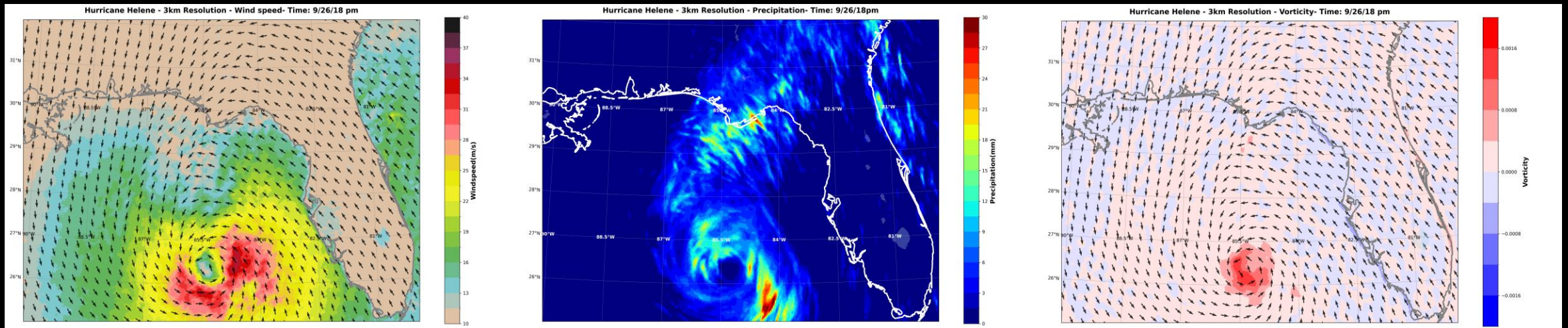


EARTH-2 CORRECTION DIFFUSION NIM USER GUIDE

A tutorial for generating AI weather forecast using NVIDIA Earth2Studio CorrDiff NIM 1.0.0



WHAT IS EARTH-2 CORRECTION DIFFUSION NIM

- NVIDIA NIM provides containers to self-host GPU-accelerated inferencing microservices for pretrained and customized AI models.
- Correction Diffusion (CorrDiff) is a neural network model that downscales surface and atmospheric variables to improve accuracy and resolution of weather data.
- The CorrDiff NIM provides a fast, accurate model behind a consistent API for producing ensembled, downsampled forecasts over the United States.



IMPORTANT PREREQUISITES

- CorrDiff NIM requires quite a bit of GPU power
- In higher education settings, we recommend gaining GPU access through a NRP Nautilus namespace [here](#) (step 7)
- After getting a namespace, admins can manage the namespace and users can gain GPU cluster access via kubernetes [here](#).

Optimized configuration

The GPU Memory and Disk Space values are in Gb; Disk Space is for both the container and the model; Profile is for what the model is optimized.

GPU	GPU Memory (GB)	Precision	# of GPUs	Disk Space
B200	192	FP32	1	64
H100	80	FP32	1	64
A100	40 & 80	FP32	1	64
RTX A6000	48	FP32	1	64
L40S	48	FP32	1	64

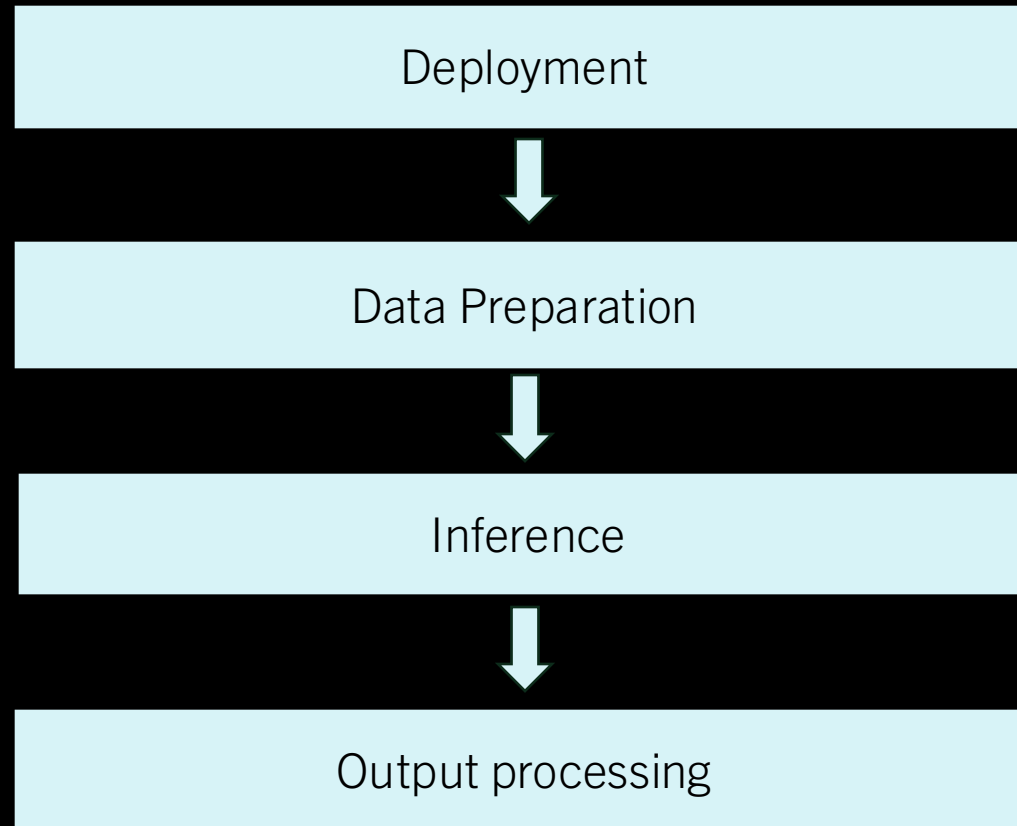
Non-optimized configuration

The GPU Memory and Disk Space values are in Gb; Disk Space is for both the container and the model.

GPU	GPU Memory (GB)	Precision	# of GPUs	Disk Space
Any NVIDIA GPU with sufficient GPU memory and compute capability ≥ 8.0	40	FP32	1	64

For models with lower GPU memory, some additional configuration may be required during runtime. Namely, the adjustment of the target sample batch size of the NIM. See the [configuration documentation](#) for additional details.

CORRDIFF NIM WORKFLOW

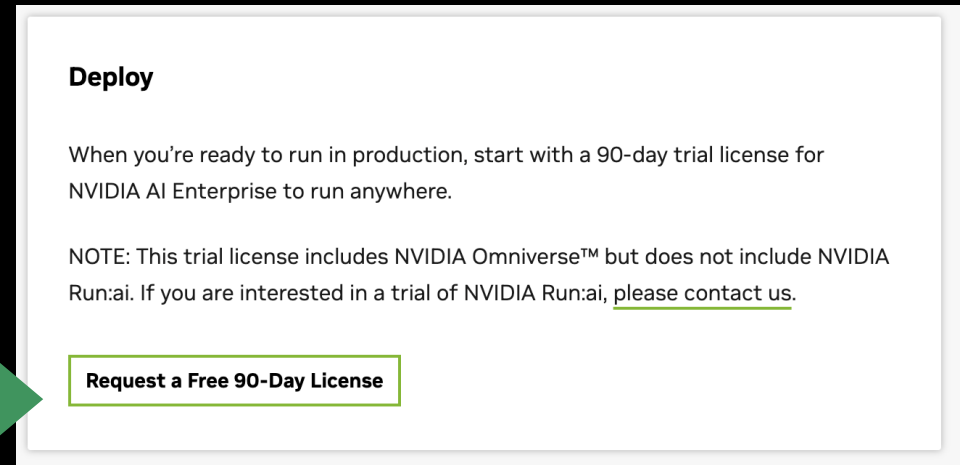


DEPLOYMENT

Pull and launch the CorrDiff NIM container, which hosts the pre-trained model and provides an HTTP API for inference.

NGC API KEY

- To acquire the NIM, we need to create an NVIDIA NGC account, register an account with NVIDIA [here](#)
- NVIDIA currently allows a 90-day trails for NVIDIA AI Enterprise, which will generate an API key that connects the NIM, users can get the free trail by clicking on [Request a Free 90-Day License](#)



NVIDIA Enterprise Account Registration

Select NVIDIA Cloud Account (NCA)

NGC Product Details

NVAIE

Included Subscriptions

- NVIDIA AI Enterprise
- Private Registry
- Omniverse
- Nvidia Runai Selfhosted
- Nvidia Runai Saas

NVIDIA Cloud Account (NCA)

"View Eligible NVIDIA Cloud Accounts" displays all the accounts where you are the owner. Choose an eligible account(linked to NGC Org) from this list to activate the product in the account or Click "Create new NCA" for a new account. "View Other NVIDIA Cloud Accounts" displays all the accounts where you are a member; you will need to reach out to your account owner to activate the product in these accounts.

- ☒ View Eligible NVIDIA Cloud Accounts ⓘ
- ☐ View Other NVIDIA Cloud Accounts ⓘ

Search accounts by name or organization

Account Name	Org Name	Subscriptions
☐ San Diego State SCIL (0936567751879124)	San Diego State SCIL	Owner Nim Dev Omniverse Dev Nvidia Dev

Create new NCA

Continue

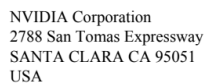
NVIDIA Enterprise Account Registration

Thank you for requesting a NVIDIA AI Enterprise evaluation entitlement. We will send you an email with the next steps within 24-48 business hours.

TIP:

Make sure you are registering with your school/corporate email instead of your personal email.

You should get an email about the entitlement certificate within 24 hours



HOW TO USE THIS CERTIFICATE

Please refer to your NVIDIA AI Enterprise Quick Start Guide for information on how to get started, including additional instructions on how to register for your entitlement.

Sales Type: EVAL

1. After you have successfully registered, you will receive an email from NVIDIA application hub to access NVIDIA Licensing Portal, NVIDIA Enterprise Support and NVIDIA NGC.
2. In order to access your entitlements, please [login](#) and bookmark this site for future access/reference.

New NVAIE 90-day Trial Participants:

1. Use the [NVIDIA Developer Forums](#) for help during your 90-day trial license for help and guidance.
2. For issues with the NVIDIA Licensing Portal or downloading components from the NGC Catalog, please open a case via the [Support Request Web form](#).
3. Contact the Customer Success team at customersuccess@nvidia.com for additional support.

Existing NVIDIA Customers (customers with existing NVIDIA entitlements):

1. For support issues, please submit support requests through the [NVIDIA Enterprise Support Portal](#).
2. Connect with your Solutions Architect for design, implementation, or application development guidance.
3. If you are unsure of who your Solutions Architect is, please contact the Customer Success team at customersuccess@nvidia.com for further assistance.

Rights and restrictions on the use, transfer and copying of the Software are set forth in NVIDIA's End User License Agreement.

NVIDIA Organization
Xinyi Hu
San Diego State Univers...

- Dashboard
- Audit
- Organization Profile
- Teams
- Usage
- Users
- Activate Subscription
- Subscriptions
- Service Keys
- External User Groups
- Webhook Management
- Enterprise IdP

NVIDIA Private Registry				Launch
Status	Term Date	Auto Renewal	Type	
Active	2026-07-24	No	Evaluation	

NVIDIA Run:ai SaaS				Launch
Status	Term Date	Auto Renewal	Type	
Active	2026-07-24	No	Evaluation	

NVIDIA AI Enterprise Essentials				Launch
Status	Term Date	Auto Renewal	Type	
Active	2026-07-24	No	Evaluation	

NVIDIA Run:ai Self-Hosted				Launch
Status	Term Date	Auto Renewal	Type	
Active	2026-07-24	No	Evaluation	

NVIDIA Developer Program				Launch
Status	Term Date	Auto Renewal	Type	
Active		No	Developer Program	

NVIDIA Omniverse Enterprise Essentials				Launch
Status	Term Date	Auto Renewal	Type	
Active	2026-07-24	No	Evaluation	

NVIDIA Developer Program				Launch
Status	Term Date	Auto Renewal	Type	
Active		No	Developer Program	

NGC Organization v0.202.17-r7

Organization

Dashboard

Audit

Organization Profile

Teams

Usage

Users

Activate Subscription

Subscriptions

Service Keys

External User Groups

Webhook Management

Account Settings

Manage your profile, preferences and credentials.

Review or edit Email Preferences then Save to commit settings

Profile

Name

Xinyi Hu

Date Created 01/16/2026

Last Login 05/10/2026 10:04 PM

Starfleet ID 24ZHI4wyPdH3EF9lcY-tzghmhSs5nlXsvsxMckddqdM

Roles

NVIDIA AI ENTERPRISE VIEWER

OMNIVERSE VIEWER

ORG OWNER

PUBLIC API ENDPOINTS USER

REGISTRY ADMIN

RUNAI SAAS ADMIN

USER ADMIN

Xinyi Hu

San Diego State University Climate Informatics Lab / corrdiff

Contact Admin

Account Settings

Setup

Organization

Terms of Use

Privacy Policy

Sign Out

xhu3914@sdsu.edu



Keys & Secrets

API Keys

Generate your own API key in order to use the NGC service through the Docker client or through NGC CLI for **Secrets Manager**, **NGC Catalog**, **Public API Endpoints**, and **NVIDIA Private Registry**.

Generate API Key

Secure Share

Securely share classified information with encrypted, ephemeral sharing links.

Share Secure Link

API Keys

[+ Generate Personal Key](#)

Personal Keys

2 Results [View](#)

Key ID	Key Name	Expiration	Key Value	Status
fac52ff0-aa10-461e-88a8-2d...	NVIDIABuild-Autogen-00	10/24/2026	nvapi-*****tWR	ACTIVE
dc1a84a3-7a3d-40bf-9139-8...	corrdiff	Never Expire	nvapi-*****70V	ACTIVE

25 Items

<< < 1 > >>

Go to

- Once the key is generated copy and paste the key that only you have access to.
- NGC does not save your key values for you
- If you ever lose your key, you can always generate a new API Key.
- Your personal key should be kept private at all times.

- With your API key values, go to your terminal and create a secret using kubectl for your key

```
kubectl create secret docker-registry ngc-  
secret-<USERNAME> \
```

```
--docker-server=nvcr.io \
```

```
--docker-username='$oauthtoken' \
```

```
--docker-password= 'YOUR API KEY' \
```

```
-n <NAMESPACE>
```

- Then create the ngc-api key

```
kubectl create secret docker ngc-api-key-  
<USERNAME> \
```

```
--from-literal=NGC_API_KEY= 'YOUR API KEY' \
```

```
-n <NAMESPACE>
```

Replace 'YOUR API KEY' with your actual key values and replace with your username.

Check that your secret and api key exist:

```
kubectl get secrets -n <YOUR NAMESPACE>
```

If they are properly stored in your system, you should see:

NAME	TYPE
ngc-secret-<USERNAME>	kubernetes.io/dockerconf
ngc-api-key-<USERNAME>	Opaque



LAUNCHING THE NIM

- To launch the NIM through kubernetes, users need to download the yaml file for deployment [here](#).
- Change <username> and <namespace> to users' unique usernames and designated namespace

- Then run the following lines in your terminal

```
kubectl create -f corrdiff-nim-deployment-  
<YOUR NAME>.yaml -n <YOUR  
NAMESPACE>
```

```
kubectl get pods -n <YOUR NAMESPACE> -  
w | grep corrdiff
```

WHAT TO EXPECT

It takes about 5-10 mins for the Kubernetes cluster to download the images(~26GB), keep checking the live status of the container in watch mode. The container is ready when the status shows RUNNING. To check your live connection logs, run `kubectl logs -f deployment/corrdiff-nim-<USERNAME>-n <NAMESPACE>` in your terminal.

```
[laurahu@Lauras-MacBook-Air ~ % kubectl get pods -n sdsu-shen-climate-lab -w | grep corrdiff  
corrdiff-nim-laurahu-58dddbbf4f-7tfsv          1/1      Running    1 (9d ago)    9d
```



REQUIRED PYTHON INSTALLATIONS

- Install earth2studio

```
pip install earth2studio
```

- Make sure your python version is >3.10
- Install submodules to ensure that data loads safely

```
pip install earth2studio[corrdiff]
```

```
pip install earth2studio[data]
```

- Download the lat and lon grid file [here](#)

WHAT YOU NEED TO KNOW ABOUT CORRDIFF NIM

What you need to know before generating weather predictions using python

CORRDIFF US ARCHITECTURE

- CorrDiff is a downscaling model based on diffusion
- Takes low-resolution GEFS inputs at 0.25 degree (25km) and downscale it to 3km resolution
- Trained on GEFS-HRRR data, so it mimics HRRR-like output
- All input and output formats are in NumPy
- The output coordinates can be obtained from the corrdiff_output_lat.npy and corrdiff_output_lon.npy
- The predictions are in 3-hr increments and the prediction limit is 24hrs

CORRDIFF NIM OUTPUTS

Output variables: [u10m, v10m, t2m, tp, csnow, cicep, cfrzr, crain]

Numerical variables:

- u10m (east to west) and v10m (north to south): windspeed at 10m height (m/s)
- t2m: air temperature at 2m height (K)
- tp: total precipitation rate (mm/hr)

Categorical variables (between 0-1):

- csnow: chances of snow
- cicep: chances of ice pellets
- cfrzr: chances of freezing rain
- crain: chances of rain

DATA PREPARATION INFERENCE & OUTPUT PROCESSING

Let's look at an example: [HERE](#)